# Sri Lanka Institute of Information Technology

## Artificial Intelligence and Machine Learning - IT2011

## Group Assignment

## Bias and Ethics in AI

**Academic Year: 2025**
**Year 02 - Semester 01**

| 2025-Y2-S1-MLB-B12G2-10 | |
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# 1. Introduction and Problem Statement

Artificial Intelligence and Machine Learning are rapidly changing 21st century education, offering predictive analytics, personalized learning, and early interventions for students at risk. However, as AI continues to be integrated into decision-making in education, there are greater concerns regarding the bias and ethical implications.

This project aims to Predict student success through a performance index while simultaneously examining bias and ethical issues in the modeling process. Specifically, we will:

- Predict **performance index** for packing students' schedules using ML models considering their study habits, previous scores, hours of sleep, extra-curriculars, and their practice patterns.

- Compare a variety of ML models to select the best performing, most accurate and most interpretable model.

- Explore possible biases in the construction of our dataset, as well as in the output of the ML models.

- Discuss the best ethical practices in the implementation of AI in education in respect to fairness, accountability, transparency and privacy.

The problem statement emphasizes that AI can be done responsibly: it is good to predict accurately, but it is also important for those predictions to be fair, to avoid discrimination.

## 2. Dataset Description

The data set used is from Kaggle: https://www.kaggle.com/datasets/nikhil7280/student-performance-multiple-linear-regression

It contains 10,000 records with six columns.

| Features | Type | Description |
|---|---|---|
| **Hours Studied** | Numeric | Daily study hours per student. |
| **Previous Scores** | Numeric | Academic scores from prior exams. |
| **Extracurricular Activities** | Boolean | Participation in non-academic activities. |
| **Sleep Hours** | Numeric | Average daily sleep duration. |
| **Sample Question Papers Practiced** | Numeric | Number of practice questions completed. |
| **Performance Index** | Numeric | Target variable (Final performance score, 0% - 100%) |

**Key Characteristics of the Dataset:**

- No missing values
- Balanced distributions across numeric features
- Possibly biased: A slight imbalance in extracurricular activity attendance (49% True, 51% False)
- Mean Performance Index: 55.2
- Standard deviation: 19.2

The data set includes both academic and non-academic contributions to student performance and lends itself to the examination of holistic contributions to predicting outcomes.

## 3. Preprocessing & EDA

Before developing machine learning models, it is important to ensure the dataset is clean, consistent, and analysis ready. The preprocessing tasks completed in this project included addressing missing entries, manipulating categorical variables, outlier removal, feature engineering, feature selection, normalization/scaling. Additionally, preprocessing contributes to improved model performance, interpretability, and fairness.

**Preprocessing Tasks:**

| Member ID | Name | Preprocessing Technique |
|---|---|---|
| **IT24103558** | Jayawickum H.W.M. | Handling Missing Values |
| **IT24103553** | Jayarathne K.P.S.U | Encoding Categorical Variables |
| **IT24102613** | Amarasinghe A.B.E | Outlier Detection & Removal |
| **IT24103506** | Siriwardana S.A.D.V.I | Feature Engineering |
| **IT24103519** | Kalubowila G.N | Feature Selection |
| **IT24103609** | Thanusikan M | Normalization / Scaling |

**1. Missing Values**

- There were no missing entries in the dataset, thus there was no need for imputation.
- The data validation confirmed that all 10,000 records were complete, which ensures a reliable model training process.

**2. Encoding Categorical Variables**

- The feature "Extracurricular Activities" is a categorical feature (Boolean, true/false).
- This is now defined numerically - true = 1, false = 0.

### 3. Outlier Detection & Removal

- Outliers were identified by utilizing the interquartile range (IQR) and through visual inspection using boxplots.
- Extreme outliers within the features of "Hours Studied," "Previous Scores," and "Performance index" were removed to help reduce skewness and improve model accuracy.

### 4. Feature Engineering

- New Feature was created to enhance predictive performance:

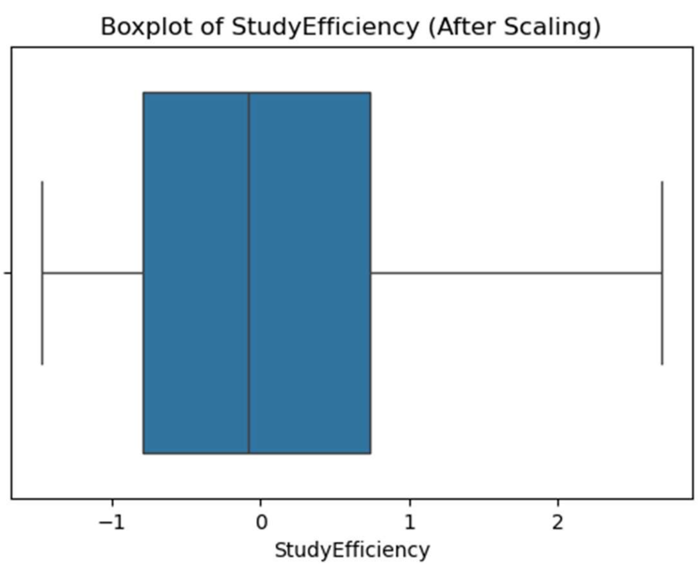
✓ ***Study Efficiency = Hours Studied / (Sleep Hours + 1)***

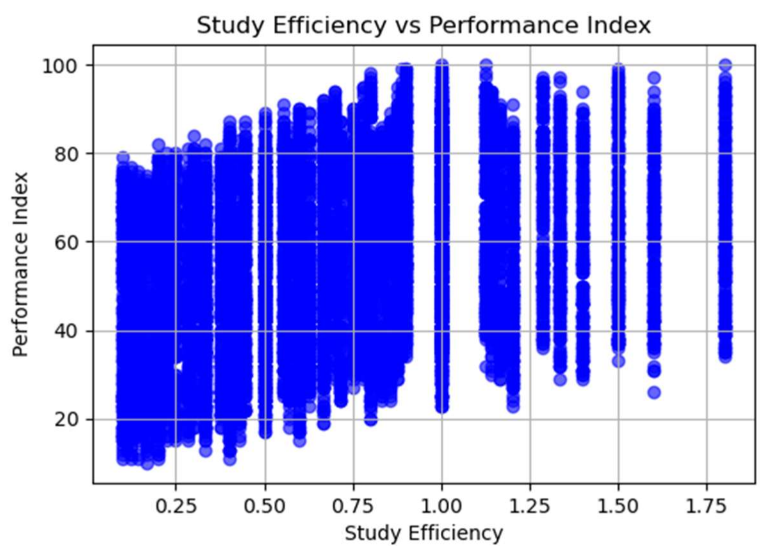
### 5. Feature Selection

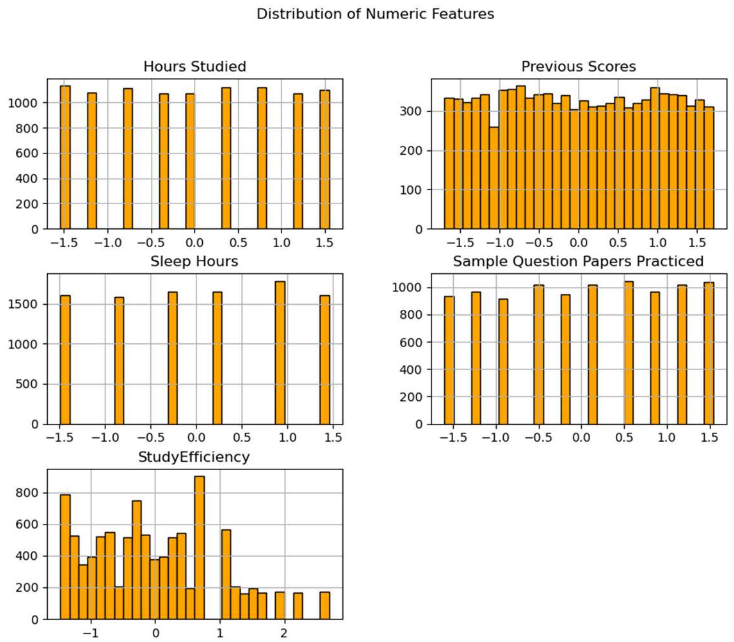
- Correlation analysis and variance thresholding were applied to identify the most relevant predictors.
- Features highly correlated with the target (Performance Index) were retained, while low importance features were dropped to reduce noise and bias.

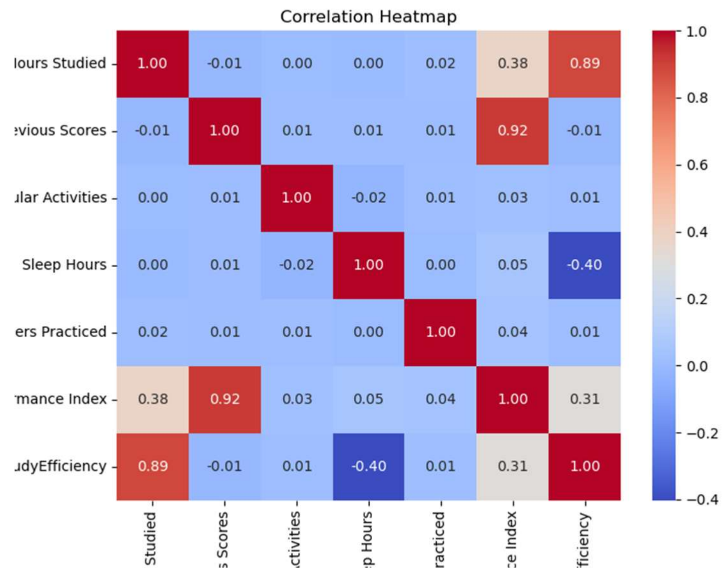
### 6. Normalization / Scaling

- Numerical features were scaled using Min-Max normalization to ensure all values are  within the range. This helps model converge faster and prevent features with larger range from dominating predictions.






## 4. Model Design and Implementation

| Member | Model | Hyperparameters | Scaled Features |
|---|---|---|---|
| **IT24103506** | Multiple Linear Regression (MLR) | MLR is used with default parameter. | Yes |
| **IT24103519** | Random Forest Regressor (RF) | n_estimators=300, max_depth=12, max_features='sqrt' | No |
| **IT24103558** | Support Vector Regression (SVR) | C=100, gamma=0.1, kernel='rbf' | Yes |
| **IT24103553** | Gradient Boosting Regressor (GBR) | n_estimators=500, learning_rate=0.05, max_depth=4 | No |
| **IT24102613** | Decision Tree Regressor (DTR) | max_depth=8 | No |
| **IT24103609** | Multi-Layer Perceptron (MLP) | hidden_layer_sizes=(100,50), max_iter=500, activation='relu' | Yes |

## Justification:

- **Multiple Linear Regression (MLR) -** Simple, interpretable and effective for linear relationships.
- **RF & DTR –** Capture non-linear interactions and can handle feature importance ranking.
- **SVR & MLP –** Suitable for non-linear patterns and complex relationships.
- **GBR –** Robust ensemble method for regression with high accuracy potential. (Keeps predictions accurate despite noisy or outlier data.)

## 5. Evaluation and Comparison

| Member | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| **IT24103506** | **Multiple Linear Regression** | **1.646589** | **2.075276** | **0.988428** |
| IT24103609 | Multi-Layer Perceptron | 1.670672 | 2.102411 | 0.988123 |
| IT24103558 | Support Vector Regression | 1.680522 | 2.122014 | 0.987901 |
| IT24103553 | Gradient Boosting Regressor | 1.703361 | 2.141791 | 0.987674 |
| IT24103519 | Random Forest Regressor | 1.951099 | 2.434481 | 0.984075 |
| IT24102613 | Decision Tree Regressor | 2.043482 | 2.575525 | 0.982176 |

## Insights:

- Multiple Linear Regression performed best overall with the lowest MAE and RMSE.
- Ensemble models (RF, GBR) and neural network (MLP) also performed strongly.
- Simple decision tree underperformed due to overfitting on some branches.
- SVR showed comparable performance to Multiple Linear Regression but slightly higher MAE.

✓ **Selected Model: *Multiple Linear Regression***

- Offers the best trade-off between interpretability and accuracy.
- Provides clear insights into how each feature contributes to students' performance.

**Special Note:**

- After evaluating several regression models, **Multiple Linear Regression (MLR)** was chosen over **Ridge Regression** for the final model. Although Ridge is a regularized form of linear regression designed to reduce the risk of overfitting, both models produced nearly identical MAE, RMSE, and $R^2$ metrics.

- The minimal difference in accuracy, given the similarity in metrics, indicates that the dataset did not consider **multicollinearity** or **coefficient instability**, which are the main issues Ridge aims to address.

- Therefore, **Multiple Linear Regression** was selected for the final model based on its **ease of interpretation** and **computational simplicity** as the best way to explain how each feature contributes to student performance without imposing additional complexity.

## 6. Ethical Considerations and Bias Mitigation

AI models in education must be designed responsibly to ensure fairness and avoid unintended harm. In our project, we carefully considered potential sources of bias and implemented strategies to uphold ethical standards.

**Sources of Bias in the Dataset:**

- **Data Bias:** Historical academic scores may reflect systemic inequalities, such as socioeconomic disparities or unequal access to resources.

- **Sampling Bias:** Certain student groups could be underrepresented, which might cause the model to perform unevenly across the population.

- **Labeling Bias:** The target variable, Performance Index, could reflect grading practices rather than true student ability, introducing subjectivity.

- **Algorithmic Bias:** Choices in model selection or parameters can favor dominant patterns in the data, potentially disadvantaging minority cases.

**Consequences of Bias:**

Bias in AI can have serious consequences, including:

- **Individual Harm**: Mislabeling or underestimating a student's potential may affect academic guidance or opportunities.
- **Societal Harm:** Biased predictions can reinforce existing educational inequalities.
- **Loss of Trust:** Students, Parents and educators may lose confidence in AI based tools if they appear unfair.

**Ethical Principles Applied:**

To ensure responsible AI use, we adhered to key ethical principles:

- **Fairness –** Models were designed to avoid discrimination based on non-academic attributes such as participation in extracurricular activities.
- **Transparency and Explainability** – We prioritized Multiple Linear Regression for its interpretable coefficients, making it easy to understand how each feature affects the prediction.
- **Accountability –** All preprocessing, model training, and evaluation steps were clearly documented.
- **Privacy and Data Protection** – Student data was anonymized handled following best practices to ensure confidentiality.
- **Human Oversight** – Predictions are advisory for educators, not automated decisions.
- **Beneficence** – Model designed to assist students and improve learning outcomes.

- By selecting Multiple Linear Regression and carefully preprocessing the dataset, we addressed bias while maintaining transparency and interpretability.

**Bias Mitigation Strategies:**

- Balanced dataset sampling to represent diverse student groups.
- Careful feature selection to prevent overemphasis on sensitive attributes.
- Explainable AI methods to monitor predictions for fairness and reliability.

## 7. Reflections and Lessons Learned

**Reflection on the Team:**

- Collaborative efforts were relied upon throughout the work to preprocess the datasets, engineer new features, and select the best model to apply.
- Achieving a balance between accuracy and interpretability was a challenge but important for an ethical AI that meets the high standards of ethical practice.
- Being attuned to biases as a part of exploring datasets provided direction for preprocessing and the selection of the model.
- Documenting all steps allowed for accountability and reproducibility.

**Lessons Learned:**

- One should always check datasets for unseen biases prior to moving to the modeling stage.
- One can get equal or better performance with simpler models than with complex ones in performance and interpretability with Multiple Linear Regression (MLR) models.
- Ethical AI requires continuous human oversight and engagement, even when models perform well.
- Collaboration gives technical and ethical lens to the goal of trustworthiness in models.

**Future Improvements:**

- To further reduce bias, collect additional socio-demographic data.
- Consider fairness aware algorithms to improve equitable and representative predictions.
- Utilize real-world feedback from educators to further improve the performance index.

## 8. References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